

Technical Guide: Troubleshooting Rotating Equipment Vibration Issues

1) Purpose

Provide a practical, repeatable method to diagnose and correct abnormal vibration in rotating equipment (motors, pumps, fans, compressors, turbines, gearboxes).

2) Safety and Preparation

- Follow LOTO and site safety procedures.
 - Confirm machine operating state (load, speed, process conditions).
 - Collect baseline data before changing anything.
 - Verify sensor mounting and calibration first (bad data is common).
-

3) Minimum Data to Collect

- Overall vibration (RMS velocity, acceleration, displacement as applicable).
 - FFT spectrum (frequency peaks, sidebands, harmonics).
 - Time waveform (impacting, modulation).
 - Phase (for balance/alignment confirmation).
 - Running speed (RPM), bearing type, blade count, gear tooth count.
 - Process data (flow, pressure, temperature, suction/discharge conditions).
-

4) Diagnostic Workflow (Recommended)

- 1. Confirm severity**
Compare against alarm/trip limits or ISO 10816/20816 guidance.
 - 2. Check operating dependency**
Does vibration change with speed, load, or flow?
 - 3. Identify dominant frequencies**
Relate peaks to $1 \times \text{RPM}$, $2 \times \text{RPM}$, bearing defect frequencies, gear mesh, vane pass, electrical frequencies.
 - 4. Use waveform + phase to narrow cause**
FFT alone is not enough for final diagnosis.
 - 5. Inspect mechanical condition**
Foundation, fasteners, soft foot, alignment, balance, bearings, coupling, looseness.
 - 6. Correct one variable at a time**
Re-test after each action and document results.
-

5) Common Fault Signatures

Symptom	Typical Signature	Likely Causes	First Actions
High 1× RPM	Strong 1×, stable phase	Unbalance	Check buildup, runout, balance condition
High 2×/3× RPM	Harmonics, axial vibration	Misalignment, bent shaft	Verify coupling alignment, check soft foot
Broad-band high-frequency	Raised noise floor, impacting	Bearing wear/lubrication issues	Lubrication check, bearing condition monitoring
Subharmonics / many harmonics	Nonlinear waveform	Mechanical looseness	Tighten supports, inspect base/grout
Peak near natural frequency	Large amplitude near specific speed	Resonance/structural issue	Bump test, stiffness/damping changes
Sidebands around gear mesh	GMF ± shaft frequency	Gear wear/eccentricity	Inspect gears, backlash, lubrication
Vane/blade pass peak	At vane/blade pass freq	Hydraulic/aerodynamic excitation	Check flow regime, recirculation/cavitation
Electrical frequency components	1× line freq / slip-related	Motor electrical faults	Motor current analysis, rotor/stator checks

6) Troubleshooting by Equipment Type

Pumps

- Check for cavitation, recirculation, air ingress, NPSH margin.
- Confirm operation near BEP (Best Efficiency Point).
- Inspect impeller damage/fouling and wear ring clearances.

Fans/Blowers

- Look for buildup on blades and uneven mass distribution.
- Check damper position and stall/surge regions.
- Verify belt condition/tension (if belt-driven).

Motors

- Separate mechanical vs electrical sources using coast-down and current signature.
- Inspect rotor bars, eccentricity, and mounting rigidity.

Gearboxes

- Trend gear mesh frequency and sideband growth.
 - Check lubrication quality, contamination, and alignment.
-

7) Corrective Action Priority

1. **Instrumentation/data quality**
 2. **Simple mechanical checks** (looseness, soft foot, fasteners)
 3. **Alignment**
 4. **Balance**
 5. **Bearing and lubrication**
 6. **Structural/resonance modifications**
 7. **Process condition optimization**
-

8) Acceptance and Verification

- Re-measure under the same operating conditions used for baseline.
 - Confirm reduction in:
 - Overall vibration levels
 - Fault-related spectral components
 - Temperature/noise anomalies
 - Keep before/after plots and maintenance records for trend history.
-

9) Recommended Documentation Template

- Asset ID, date/time, operating point
- Symptoms observed
- Measurement locations and sensor types
- Key FFT/waveform/phase findings
- Root cause conclusion
- Corrective action performed
- Post-repair verification data
- Follow-up date and trend status